

Project Design Phase

Solution Architecture

Date	29/09/2026
Team ID	SWTID-2026-4654
Project Name	AI StudyBuddy - AI assistant for smarter learning and study support
Maximum Marks	4 Marks

Solution architecture bridges the gap between the problem and the technical solution. AI StudyBuddy follows a modular RESTful architecture: requests pass through Express, routes and authentication middleware before controllers call the database or the AI service.

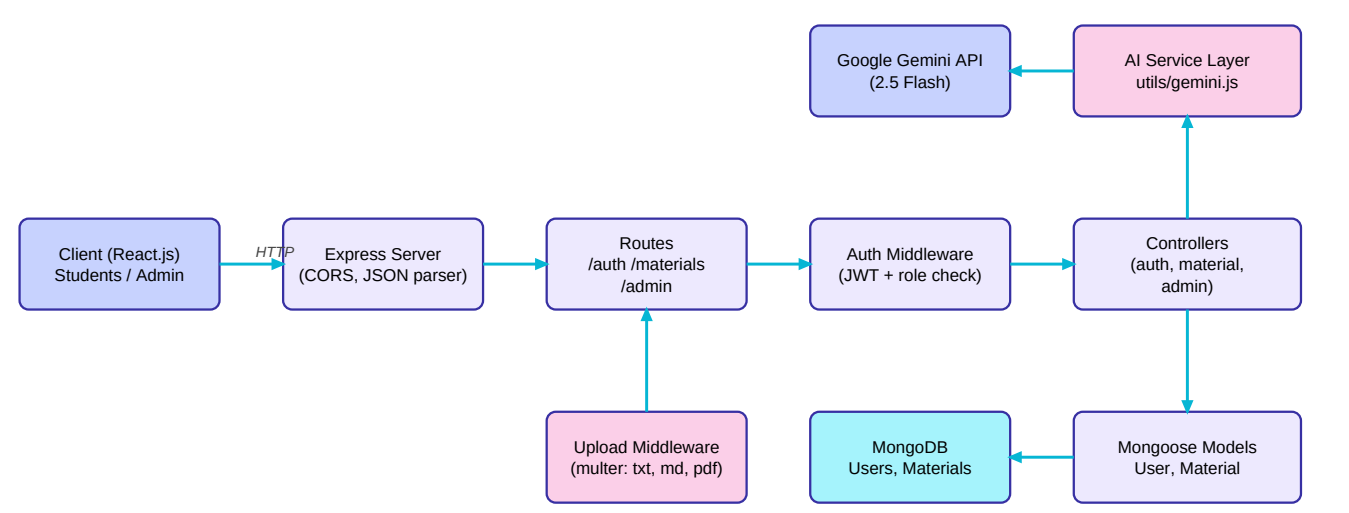


Figure 1: AI StudyBuddy architecture and data flow

Components:

- **Client Application:** a React.js frontend where users register, log in, upload material and use the AI features through REST APIs.
- **Express Server:** receives HTTP requests, applies JSON parsing and CORS, routes them and returns responses (with a global error handler).
- **Route Handlers:** /api/auth, /api/materials and /api/admin endpoints for authentication, study materials, AI features and administration.
- **Authentication Middleware:** verifies the JWT bearer token and enforces admin-only access.
- **Upload Middleware:** multer accepts .txt, .md and .pdf files up to 5 MB.
- **AI Service Layer:** utils/gemini.js sends prompts to Google Gemini 2.5 Flash and returns summaries, flashcards, quizzes and study plans.
- **Database Layer:** Mongoose models (User, Material) in MongoDB store accounts, materials and the AI-generated resources.
- **Data model:** User to Study Material is one-to-many; each material can carry a summary, flashcards, a quiz and a study plan.